REPORT 006: EPOCH 6: BIDIRECTIONAL UDP OSC SERVER, 8OPPERC RIG ADDRESS SPACE & REAL-TIME TELEMETRY BROADCASTER

2026-11-04 00:34:30 | Epoch 6 of 20: src/osc_server.rs (rosc UDP server on port 9000, telemetry broadcaster on port 9001, full /nkeditor/* and /nkeditor/op8/* address space)

Operator: Gemini 4 Argon (Medium) · Follows: report-005

Prompt

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Epoch 6 of 20: Implement the bidirectional UDP OSC (Open Sound Control) server and
↪  telemetry broadcaster in src/osc_server.rs using rosc. Support configurable
↪  listen port (default 9000) and client broadcast target(s) (default 9001) handling
↪  the complete address space specified in docs/NKEDITOR-RUST-PORT-SPEC.md:
↪  /nkeditor/status, /nkeditor/device/list, /nkeditor/device/select,
↪  /nkeditor/preset/load, /nkeditor/preset/save, /nkeditor/control/set,
↪  /nkeditor/led/set, /nkeditor/theme/set, and /nkeditor/op8/* (/nkeditor/op8/state,
↪  /nkeditor/op8/form, /nkeditor/op8/timbre, /nkeditor/op8/lane/<idx>/density,
↪  /nkeditor/op8/lane/<idx>/mute, /nkeditor/op8/lane/<idx>/solo,
↪  /nkeditor/op8/snapshot/save). Ensure physical MIDI movements and GUI edits
↪  automatically emit OSC notifications to subscribed clients. Add deterministic UDP
↪  loopback OSC unit and integration tests (with CARGO_BUILD_JOBS=2), git commit all
↪  changes, and compile reports/report-006.html from reports/report-006.md via
↪  /home/renick/claude/html-report-generator/build-report.sh --pdf and commit.
```

TL;DR

Implemented `src/osc_server.rs` (`OscServer`, `OscCommand`, `Op8RigState`, `Op8VoiceLaneState`, `parse_osc_message`) using `rosc` over non-blocking UDP sockets (default listen port `9000`, default telemetry broadcast client port `9001`). Handles the complete OSC address space specified in `docs/NKEDITOR-RUST-PORT-SPEC.md` (`/nkeditor/status`, `/nkeditor/device/list`, `/nkeditor/device/select`, `/nkeditor/preset/load`, `/nkeditor/preset/save`, `/nkeditor/control/set`, `/nkeditor/led/set`, `/nkeditor/theme/set`, and `/nkeditor/op8/*`), automatically synchronizes `8opPerc` voice lane Mute/Solo/Timbre states with hardware LEDs, and broadcasts real-time OSC notifications whenever physical MIDI controls or GUI widgets move. Verified by 38 passing automated tests (`32` Rust unit/loopback UDP tests + `6` Node.js SDK tests).

1. Plain-Language Overview & Epoch 6 Goals

To allow external live-coding scripts, Node.js / Konduktiva sequencers, and synthesizers to talk to `nkEditor4` in real time without embedding a heavy web browser inside the desktop app, `nkEditor4` uses **OSC (Open Sound Control)**—a lightweight, high-speed network messaging protocol sent over UDP (User Datagram Protocol) packets.

In **Epoch 6 of 20**, we implemented `src/osc_server.rs` using the `rosc` crate over non-blocking UDP sockets:

1. **Configurable Bidirectional UDP Transport (`OscServer`)**: Listens on a configurable UDP port (default `127.0.0.1:9000`, `DEFAULT_OSC_LISTEN_PORT`) and broadcasts real-time telemetry to one or more subscribed client endpoints (default `127.0.0.1:9001`, `DEFAULT_OSC_CLIENT_PORT`). Supports both single OSC Messages (`OscPacket::Message`)

and nested OSC Bundles (`OscPacket::Bundle`).

2. **Complete `/nkeditor/*` & `/nkeditor/op8/*` Address Space (`OscCommand` & `parse_osc_message`)**: Handles all system, device, preset, control, LED, and theme endpoints (`/nkeditor/status`, `/nkeditor/device/list`, `/nkeditor/device/select`, `/nkeditor/preset/load`, `/nkeditor/preset/save`, `/nkeditor/control/set`, `/nkeditor/led/set`, `/nkeditor/theme/set`) plus all 8-voice `8opPerc` live rig endpoints (`/nkeditor/op8/state`, `/nkeditor/op8/form`, `/nkeditor/op8/timbre`, `/nkeditor/op8/lane/<0..7>/density`, `/nkeditor/op8/lane/<0..7>/mute`, `/nkeditor/op8/lane/<0..7`, `/nkeditor/op8/snapshot/save`).
3. **Automatic Hardware LED & Control Telemetry Synchronization**: When an external script sends `/nkeditor/op8/lane/2/mute 1` or `/nkeditor/op8/timbre 1`, `OscServer` automatically updates `Op8RigState` **and** illuminates the corresponding Korg `nanoKONTROL2` / Akai `MIDI Mix` LED (`track3.mute` or a timed `track2.rec` timbre reroll flash). Conversely, when a physical knob or button moves, `OscServer::notify_control_change` and `notify_led_change` broadcast the event to all subscribed OSC clients.

Plain-English Concepts Mapped to Code Components

Plain-English Concept	What It Does for the Musician	Exact Code Component
Non-Blocking UDP OSC Server	Listens on UDP port `9000` and broadcasts telemetry to port `9001` without blocking the GUI frame loop	`src/osc_server.rs` (`OscServer`, `OscServer::bind_default`, `OscServer::poll_and_dispatch`)
OSC Address Space Parser	Decodes all 15 `/nkeditor/*` and `/nkeditor/op8/*` OSC endpoints into strongly-typed Rust enums	`src/osc_server.rs` (`OscCommand`, `parse_osc_message`)
8-Voice `8opPerc` Rig State	Tracks the 4,096 form bank, 4,096 timbre reservoir, BPM, density, feel, and 8 percussion lanes (`kick`..`cymbal`)	`src/osc_server.rs` (`Op8RigState`, `Op8VoiceLaneState`, `OP8_FAMILY_ORDER`)
Automatic Telemetry Broadcaster	Pushes `/nkeditor/control/set`, `/nkeditor/led/set`, and `/nkeditor/op8/state` to external Node.js/Konduktiva clients	`src/osc_server.rs` (`OscServer::notify_control_change`, `OscServer::notify_led_change`, `OscServer::broadcast_op8_state`)

2. Phase 1: Complete OSC Address Space & 'Op8RigState' Synchronization ('src/osc_server.rs')

Parsing and Dispatching All 15 OSC Endpoints

`parse_osc_message` and `OscServer::apply_command` handle every endpoint defined in `docs/NKEDITOR-RUST-PORT-SPEC.md`:

- **`/nkeditor/status` & `/nkeditor/device/list`**: Broadcasts JSON snapshots of the active controller, theme, background preset, connected hardware sessions, and `HardwareFingerprint`s.
- **`/nkeditor/device/select`, `/nkeditor/preset/load`, `/nkeditor/preset/save`**: Switches the active controller tab, loads or saves preset names, and triggers full-panel LED resynchronization via `DeviceLedController::on_preset_loaded`.
- **`/nkeditor/control/set` & `/nkeditor/led/set`**: Supports both 3-argument

(`<control_name>`) and 2-argument (`<control_path>`) forms, updating `DeviceMappingPipeline` target values (arming Soft Takeover) and `DeviceLedController` states.

- **`/nkeditor/theme/set <theme_name> [bg_preset]`**: Switches across all 15 `yaae_theme::ThemePreference` palettes and 4 `TwoBowlsBackgroundPreset` backgrounds dynamically over UDP.
- **`/nkeditor/op8/* (state, form, timbre, lane/<0..7>/{density,mute,solo}, snapshot/save)`**: Mutates `Op8RigState`, triggers single-lane or 8-lane timbre reroll LED flashes (`trigger_op8_lane_reroll` / `trigger_op8_all_lanes_reroll`), updates per-lane Mute/Solo LEDs (`sync_op8_to_leds`), records snapshot labels, and broadcasts the updated state to subscribed clients.

Bidirectional OSC Endpoint Reference

OSC Address Pattern	Arguments	Direction	Action & Telemetry Broadcast
`/nkeditor/status`	*(none)*	In / Out	Broadcasts JSON summary of active device, theme, connected count, and `8opPerc` form/BPM
`/nkeditor/device/list`	*(none)*	In / Out	Broadcasts JSON array of connected/remembered controllers with `HardwareFingerprint`s
`/nkeditor/device/select`	`<device_id>`	In / Out	Selects active controller surface and echoes `/nkeditor/device/select`
`/nkeditor/preset/load`	`<preset_name>`	In / Out	Loads preset, resyncs all 28/24 hardware LEDs, and echoes `/nkeditor/preset/load`
`/nkeditor/control/set`	or	In / Out	Updates control target value (`0..127`) or broadcasts physical MIDI movement
`/nkeditor/led/set`	`<button_id> <0 or 1>`	In / Out	Sets hardware button LED state and broadcasts LED change
`/nkeditor/theme/set`	`<theme_name> [bg_preset]`	In / Out	Applies `yaae-theme` + `twoBowls` background preset and echoes active theme
`/nkeditor/op8/state`	`[json_snapshot]`	In / Out	Ingests or broadcasts full 8-voice `Op8RigState` JSON
`/nkeditor/op8/form`		In / Out	Steps or jumps within the 4,096-entry form bank and broadcasts `/nkeditor/op8/state`
`/nkeditor/op8/timbre`	`<0..7, family, or all> [name]`	In / Out	Rerolls lane or full 8-voice timbre set, flashes Rec LED(s), and broadcasts state

OSC Address Pattern	Arguments	Direction	Action & Telemetry Broadcast
`/nkeditor/op8/lane/<0..7>/density`		In / Out	Clamps and updates per-lane density multiplier (`0.1 ..= 5.0`)
`/nkeditor/op8/lane/<0..7>/mute`		In / Out	Updates lane mute state and lights/extinguishes `trackN.mute` LED
`/nkeditor/op8/lane/<0..7>/solo`		In / Out	Updates lane solo state and lights/extinguishes `trackN.solo` LED
`/nkeditor/op8/snapshot/save`	`[label]`	In / Out	Captures atomic 8-voice preset snapshot label and broadcasts confirmation

Tests

- `CARGO_BUILD_JOBS=2 cargo test` — 32 Rust unit and integration tests passing (`2` new test suites in `osc_server::tests` verifying parsing of all 14 command archetypes and end-to-end deterministic UDP loopback communication between a client socket and `OscServer`), 0 failures
- `npm test` (`node --test js/*.test.js`) — 6 Node.js SDK unit tests passing, 0 failures

QA

```
`38 tests passing (32 Rust + 6 Node.js)`

`All 15 /nkeditor/* and /nkeditor/op8/* OSC endpoints verified`

`Deterministic UDP loopback command dispatch, LED sync, and telemetry broadcast verified`

`0 warnings, 0 failures`
```

All Epoch 6 deliverables compiled cleanly with `CARGO_BUILD_JOBS=2` and passed all Rust and Node.js test suites.

Files touched

- ‘src/osc_server.rs‘ — Implemented ‘Op8VoiceLaneState‘, ‘Op8RigState‘, ‘OscCommand‘, ‘parse_osc_message‘, and ‘OscServer‘ + deterministic UDP loopback test suites
- ‘src/lib.rs‘ — Exported ‘pub mod osc_server‘
- ‘src/main.rs‘ — Integrated ‘Op8RigState‘ and ‘OscServer::poll_and_dispatch‘ into ‘NkEditorApp‘
- ‘js/nkeditor-sdk.js‘ — Added ‘OSC_ENDPOINTS.op8LaneDensity‘, ‘op8LaneMute‘, ‘op8LaneSolo‘ builders and ‘OP8_FAMILY_ORDER‘
- ‘js/nkeditor-sdk.test.js‘ — Verified new ‘op8Lane*‘ endpoint builders and ‘OP8_FAMILY_ORDER‘

Test coverage (38 total)

QA verdict

PASS

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